MIL-E-3677C
7 February 1985
SUPERSEDING
MIL-E-3677B(ME)
16 December 1975

MILITARY SPECIFICATION

ERECTION EQUIPMENT

FOR BRIDGE, FIXED, STEEL,

PANEL, BAILEY TYPE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers bridge-erection equipment used in erecting a portable panel, Bailey-type bridge.

1.2 Classification. Equipment shall be the following designations, as specified (see 6.2):

Bar, carrying
Extractor, pin
Jack, chord
Lever, panel
Roller, plain
Roller, rocking
Shoe, jack
Template, roller, rocking
Template, roller, plain
Tongs, carrying
Wedge

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: STRBE-DS, Fort Belvoir, VA 22060-5606, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5420

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel and Seals.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-G-20241	- Gasket Material, Wool Felt, Impregnated, Adhesive, Pressure-Sensitive.
MIL-T-22085	- Tape, Pressure-Sensitive, Adhesive, Preservation and Sealing.
MIL-S-81733	- Sealing and Coating Compound, Corrosion Inhibitive.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods.

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2.1.2 Government drawings. The following Government drawings form a part of this specification to the extent specified herein.

DRAWINGS

ME

TA13220E4774 - Erection Equipment for Panel Bridge, M2 (Bailey Type).

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASME - Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.
ANSI/AWS A5.1 - Covered Carbon Steel Arc Welding Electrodes.
ANSI/AWS D1.1 - Structural Welding Code-Steel, Section 5, Qualification.

(Applications for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The erection equipment shall be as shown on Top Assembly TA13220E4774 and as specified herein.

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3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the erection equipment. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available upon request, for inspection by the contracting officer or the designated representative.

3.2 First article. When specified (see 6.2), a sample of each erection equipment component shall be subjected to first article inspection (see 4.3 and 6.3).

3.3 Material. Material shall be as specified herein and as shown on the drawing. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.4).

3.3.1 Material deterioration and control. The erection equipment shall be fabricated from compatible materials, inherently corrosion and deterioration resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environment to which the item may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, shall be electrically insulated from one another to minimize or prevent galvanic corrosion. Insulation may be provided by an insulating barrier such as a corrosion inhibiting sealant conforming to MIL-S-81733 or chromate tape conforming to MIL-G-20241. Protection against any galvanic corrosion could also be obtained by exclusion of the electrolyte if feasible.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make information available, upon request, to the contracting officer or designated representative.

3.4 Welding electrodes. Welding electrodes shall conform to ANSI/AWS A5.1, classification E6010, E6011, E6012, E6013, or E6020, as prescribed for the operation.

3.5 Initial lubrication service. The erection equipment shall be serviced with standard military lubricants conforming to MIL-G-10924. Each lubricated item of equipment shall be tagged in a conspicuous place to indicate the temperature range and grades of lubricants used.

3.6 Identification marking. Each item of equipment shall be identified in accordance with MIL-STD-130 and as shown on the drawings.

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3.7 Treatment and painting. The erection equipment shall be treated and painted in accordance with MIL-T-704, type A, color forest green. The working parts of the cord jack, pin extractor, roller shafts, and bearings, shall not be painted.

3.8 Workmanship.

3.8.1 Steel fabrication. Steel used in the fabrication of the erection equipment shall be straight and free from laminations. Straightening of materials shall be done by methods that will not change the physical properties of the materials. All bends shall be made with metal dies or fixtures in order to insure uniformity of size and shape. Steel may be cut by shearing, sawing, or flame cutting. Burned surfaces of flame-cut material shall be ground or machined to remove cooling checks. All burrs shall be removed. External surfaces shall be free of sharp edges and corners except where sharp edges and corners are required or where they are not detrimental to safety.

3.8.2 Wood fabrication. All wood parts shall be free from splinters, holes, unsound knots, and other imperfections. The taper of wood parts shall be uniform for the entire length of the taper.

3.8.3 Bolted connections. Boltholes shall be punched or drilled and all burrs shall be removed.

3.8.4 Rivet and pin connections. Rivet and pin holes shall be drilled to size or punched 1/8 inch under size and reamed to size. Holes shall not have any burrs. Rivet heads shall be concentric with the rivets and shall be in full contact with the surface of the members. The pins or rivets used in rotating or pivoting joints shall allow the joints to operate without binding.

3.8.5 Welders and welding.

3.8.5.1 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall obtain certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

ANSI/ASME - Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.
ANSI/AWS D1.1 - Structural Welding Code-Steel, Section 5, Qualification.

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Contractors who make only horizontal welds need not qualify welders for "all position welding". Subject to approval by the Government, the contractor's standard welder qualification may be substituted in lieu of the above codes provided that the contractor's procedure is equivalent to the above codes. The contractor shall be responsible for determining that automatic welding equipment operators are capable of producing quality welds in accordance with ANSI/ASME or ANSI/AWS codes. The certification shall be made available for review by the contracting officer or the contracting officer's representative.

3.8.5.2 Welding. The surfaces of parts to be welded shall be free from scale, paint, grease, and other foreign matter. Welding shall be done by the shielded electric-arc method and shall conform to the ANSI/AWS D1.1, Structural welding code and to the requirements herein. Before welding over previously deposited weld metal, all traces of slag shall be removed and the deposit and the adjoining base metal shall be wire-brush cleaned. All welded parts shall be assembled in jigs or frames and welded together while held in position. The finished erection equipment shall be within toleranced dimensions. Complete and uniform penetration and fusion of the metals shall be obtained on all welds. Straightening and stress relieving shall be permitted to relieve distortion, provided the straightening process does not change the physical characteristics of the metal used in the erection equipment.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspections. The inspections specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

c. Inspection of packaging (see 4.6).

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4.3 First article inspection.

4.3.1 Examination. The first article erection equipment shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling for examination. Sampling for examination shall be in accordance with MIL-STD-105.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. AQL shall be 1.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.5 Inspection procedure.

4.5.1 Examination. The erection equipment shall be examined as specified herein for the following defects:

Major

101. Dimensions not as specified on the drawings.
102. Material not as specified.
103. Materials are not resistant to corrosion and deterioration or treated to be resistant to corrosion and deterioration for the applicable storage and operating environments.
104. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
105. Contractor does not have documentation available for identification of material, material finishes or treatment.
106. Components and parts missing or not as specified.
107. Weld defects such as flux inclusions, spatter, undercutting, porosity, or lack of fusion with parent metal.
108. Nonfunctioning of parts and components.
109. Edges not having the specified radius; or burrs on metal parts.
110. Wood parts containing splinters, knot holes, or unsound knots.
111. Workmanship not as specified.
112. Welding not as specified.
113. Welding certification not as specified.

Minor

201. Treatment and painting not as specified.
202. Identification markings incorrect or missing.
203. Screws, nuts, bolts, washers, or cotter pins missing or not as

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4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

114.	Materials and containers not as specified for level A or B. Each incorrect material or container shall constitute one defect.
115.	Unfinished or machined surfaces not coated with preservative and wrapped or covered when required, as specified for level A.
116.	Components of unlike description packed together for level A or B.
117.	Quantities packed together exceed the weight limitation of the box for level A or B.
118.	Blocking, bracing, and anchoring not in accordance with the reference document for level A or B.
119.	Preservation and packing not in accordance with the referenced document as specified for commercial.
120.	Marking missing, illegible, incorrect, or incomplete.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial, as specified (see 6.2).

5.1.1 Level A. Unpainted unfinished metal surfaces of components requiring the application of a contact preservative in accordance with MIL-P-116, shall be coated with type P-1 preservative. Exposed machine surfaces requiring the application of a contact preservative in accordance with MIL-P-116, shall be coated with type P-6 or P-11 preservative and the coated surface covered or wrapped with barrier material conforming to MIL-B-121, type I, grade A, class 2, and secured in place with tape conforming to MIL-T-22085, type II. The preservatives shall conform to the applicable specifications listed in and shall be applied in accordance with MIL-P-116.

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5.1.2 Commercial. Commercial preservation shall be in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A or B, or commercial, as specified (see 6.2).

5.2.1 Level A. Erection equipment of like description shall be packed together in close-fitting boxes conforming to PPP-B-601, overseas type, style optional, or PPP-B-621, class 2 style optional, in quantities not to exceed the weight limitation of the box. The contents shall be blocked, braced, and anchored as applicable to prevent free movement in accordance with MIL-STD-1186. Box closure and strapping shall be in accordance with the appendix to the applicable box specification. Strapping shall conform to QQ-S-781, type I or IV, size as applicable. Unless otherwise specified (see 6.2) strapping shall be finish B. When specified (see 6.2) strapping shall be finish A.

5.2.2 Level B. Packing shall be as specified in 5.2.1 for level A except boxes shall be domestic type or class as applicable and strapping shall be finish A.

5.2.3 Commercial. Commercial packing shall be in accordance with ASTM D 3951.

5.3 Marking.

5.3.1 Military. Marking for military levels of protection (level A or B) shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Marking for commercial packaging shall be in accordance with ASTM D 3951.

6. NOTES

6.1 Intended use. The erection equipment is intended for use in erecting a portable panel Bailey-type bridge.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Designation of equipment desired (see 1.2).

c. When a first article is required for inspection and approval and the number of units required (see 3.2).

d. Degree of preservation and packing required (see 5.1 and 5.2).

e. When other than finish B strapping is required (see 5.2.1).

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6.3 First article. When a first article inspection is required, the item will be tested and should be a first produced item. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, and approval of the first article.

6.4 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.3).

Custodian:
Army - ME

User activities:

DLA - CT

Preparing activity:
Army - ME

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-E-3677C	Erection Equipment for Bridge, Fixed, Steel, Panel, Bailey Type

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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